ETHAMBUTOL

Optic neuritis
Check vision monthly

CHEMICAL NAME (2S,2′S)-2,2′-(ethane-1,2-diyldiimino)dibutan-1-ol (as dihydrochloride)
CLASS First-line antituberculosis drug
TRADE NAMES Myambutol · Etibi · EMB

THERAPEUTIC USES / INDICATIONS

- Tuberculosis; intensive phase (HRZE) to prevent resistance until susceptibility known.
- *Mycobacterium avium* complex (MAC) infections (with macrolide).

ADMINISTRATION

- **PO** once daily, with or without food, usually in fixed-dose combination.

DOSAGES

- **Adult:** 15 mg/kg once daily (re-treatment: 25 mg/kg for the first 60 days, then 15 mg/kg).
- **Children:** 25 mg/kg once daily for 60 days, then 15 mg/kg (UK label); visual testing needed.
- Renal impairment (CrCl <30): 15–25 mg/kg 3 times weekly.

PHARMACOKINETICS

- **Peak:** 2–4 h. Bioavailability ~80%.
- Partly metabolised in liver.
- Excreted mainly unchanged in urine (accumulates in renal failure). **t½:** 3–4 h.

PHARMACODYNAMICS

- Inhibits **arabinosyl transferase** → impaired arabinogalactan synthesis in mycobacterial cell wall → bacteriostatic.
- Prevents emergence of resistance to other drugs.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, abdominal pain.
- Headache, dizziness.
- Hyperuricaemia.

ADVERSE EFFECTS (SERIOUS)

- Optic (retrobulbar) neuritis: ↓ visual acuity, central scotoma, red–green colour blindness; usually reversible if stopped early.
- Peripheral neuropathy.
- Hypersensitivity, thrombocytopenia (rare).

CONTRAINDICATIONS & CAUTIONS

- Pre-existing optic neuritis.
- Patients unable to report visual changes (young children, unconscious) — use with caution and monitoring.
- **Caution:** renal impairment, elderly, diabetic retinopathy.

KEY INTERACTIONS

- **Aluminium antacids:** ↓ absorption.

NURSING MANAGEMENT

- Baseline visual acuity and colour vision (Ishihara); monitor monthly.
- Stop and report any visual change.
- Check renal function for dosing.
- Monitor uric acid.
- Support adherence.

PATIENT EDUCATION

- Report blurred vision or difficulty telling red from green immediately.
- Do not stop other TB tablets.
- Have regular eye checks.

KEY POINTS

- First-line TB drug (E) — bacteriostatic.
- Baseline and monthly vision checks.
- Optic neuritis: red–green colour blindness.
- Renally excreted.

SOURCES CHECKED **[215]** Ethambutol 100 mg film-coated tablets (UK SmPC) **[216]** Ethambutol Hydrochloride, oral (US label) **[217]** Ethambutol Hydrochloride, oral (US label, STI Pharma)

Full references p. 432–457